

Reg No.: _____

Name: _____

APJ ABDUL KALAM TECHNOLOGICAL UNIVERSITY

Fifth Semester B.Tech Degree Regular and Supplementary Examination December 2023 (2019 Scheme)

Course Code: CXT307**Course Name: VIRTUAL REALITY**

Max. Marks: 100

Duration: 3 Hours

PART A*(Answer all questions; each question carries 3 marks)*

Marks

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|----|--|---|
| 1 | What is virtual reality? List out the key components of VR. | 3 |
| 2 | Write a note on perception engineering. | 3 |
| 3 | Differentiate passive input component and active input component in VR. | 3 |
| 4 | Explain four key elements of virtual reality experience. | 3 |
| 5 | Explain the concept of geometric models in object representation. | 3 |
| 6 | Explain the concept of object rotation in 3D. | 3 |
| 7 | Differentiate motions in the real world and virtual world. | 3 |
| 8 | Explain the different phases of collision detection. | 3 |
| 9 | List out the design consideration for the development of interaction mechanisms in VR. | 3 |
| 10 | Explain the key parameters and variables involved in the Doppler effect equation. | 3 |

PART B*(Answer one full question from each module, each question carries 14 marks)***Module -1**

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|----|---|----|
| 11 | a) Explain the areas of application of Virtual Reality. | 10 |
| | b) Define the term interactivity in VR. | 4 |
| 12 | a) Explain the evolution of VR systems. | 10 |
| | b) Differentiate Virtual reality and Augmented reality. | 4 |

Module -2

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|----|----|---|----|
| 13 | a) | Discuss the different ways of monitoring participant's interactions in the Virtual world. | 10 |
| | b) | Write a note on the haptic display. | 4 |
| 14 | a) | Explain aural displays in detail. | 10 |
| | b) | Explain different types user tracking and user input devices in a VR environment. | 4 |

Module -3

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|----|----|--|----|
| 15 | a) | Apply different transformation matrices to transform a square of side length unit 1 into all possible cases. Assume the coordinates accordingly. | 14 |
| 16 | a) | Write a note on perception of motion in virtual reality | 9 |
| | b) | Explain the perception of depth and perception of motion | 5 |

Module -4

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|----|----|--|---|
| 17 | a) | Apply Range-Kutta method to find an approximate value of y when x=0.2, where y=1 when x=0. (Hint: take value for h=0.1) for the equation $dy/dx=x+y$. | 8 |
| | b) | How do vestibular systems detect linear and angular motion? | 6 |
| 18 | a) | Explain the six types of vection in virtual environments, their characteristics, and the visual effects they produce. | 6 |
| | b) | Explain velocity and acceleration in one dimensional world. | 8 |

Module -5

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|----|----|--|----|
| 19 | a) | Explain the concept of locomotion in the context of VR. | 9 |
| | b) | List out the techniques used to perform object manipulation in VR. | 5 |
| 20 | a) | Explain how signal processing is employed for virtual reality. | 10 |
| | b) | Explain the concept of auralization in the context of virtual reality. | 4 |
